**UNIT WISE STUDY MODULE FOR CHEMISTRY CLASS -XI**

## CHAPTER 1 SOME BASIC OF CHEMISTRY

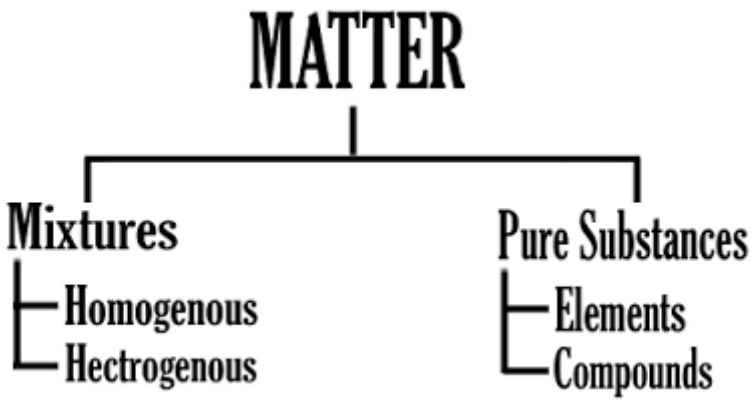
## BASIC CONCEPTS OF CHEMISTRY

Chemistry deals with the composition, structure and properties of matter.
Anything that has **mass** and occupies **space** is called matter.
Matter mainly exists in three forms - **solid**, **liquid** and **gaseous** forms. These forms are inter convertible by changing the temperature and pressure.

**SOLID <----> LIQUID <----> GAS**

- Solids have a definite volume and definite shape. The particles are closely attached and there is not much freedom of movement.
- Liquids does not have a definite shape but have a definite volume. The particles are not so close and there is a freedom of movement.
- Gases neither have definite shape nor have a definite volume. The particles are situated as far as they can and are able to move freely.

### BASIC CLASSIFICATION OF MATTER




**Mixtures** - are the combination of two or more atoms which are combined in a ratio

- If the ratio is uniform throughout the mixture, then it is a **homogeneous mixture**
- If the ratio is not uniform throughout the mixture, then it is a **heterogeneous mixture**

**Pure Substances** - consist of either one type of atoms or the combination of them in a fixed ratio.

- If only one type of atom is present in it, then it is called an **element**
- If more than one type is present, it is called a **compound**

**Molecules** are are the combination of two or more atoms

**Physical Properties** can be measured or identified without changing the composition of the substance.
**Chemical Properties** occur due to a chemical change on the constituents.

**Mass:** It is the amount of matter present in a substance. It is independent of location.
**Weight:** It is the amount of force excerted on an object by gravity. It changes according to the change in gravitational force.
**Volume:** It is the amount of space occupied by an object.

## LAWS OF CHEMICAL COMBINATION

1. **Law of Conservation of Mass (Antonie Lavoisier)** - Mass can neither be created nor be destroyed.
2. **Law of Definite Proportions (Joseph Proust)** - A given compound always contains the same elements in the same proportion by mass.
3. **Law of Multiple Proportions (John Dalton)** - When two or more elements combine to form more than one compound, a fixed mass of one compound to a different mass of other always brings simple whole number ratios.
4. **Gay Lussac's Law** - When gases combine or are produced in a chemical reaction, they do so in a simple ratio provided all gases are in the same temperature and pressure.
5. **Avogadro's Law** - Equal Volumes of gases at the same temperature and pressure should contain equal number of molecules.

## DALTON'S ATOMIC THEORY

1. Matter consist of indivisible atoms
2. All atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.
3. Compounds are formed when atoms of different elements combine in a fixed ratio.
4. Atoms are neither created or are destroyed in a chemical reaction.

**Atomic Mass** - It is defined as the mass equal to 1/12th the mass of a Carbon-12 atom. It is represented by '**amu**' (atomic mass unit) or '**u**' (unified mass).
1 amu or 1 u = $1.66056 \times 10^{-24}$

**Average Atomic Mass** - It is atomic mass of elements or compounds having isotopes. It is the sum of the product of the Relative Abundance and Atomic mass of the Isotopes.

**Molecular Mass** - It is the sum of the atomic mass of the elements present in the molecule. Eg: $CH_4$ - (1 X 12) + (4 X 1) = 16 amu

**Formula Mass** - It is used for measuring the molecular mass of the entities which does not exist in solid form. Eg: NaCl

**Avogadro Number** - It is the amount of atoms or molecules present in one mole of a substance. Avogadro Number($N_{A)}$ = $6.023 \times 10^{23}$

**Molar Mass** - Mass of one mole of a substance.

**Percentage Composition** - It is the Percentage of an element present in a compound.

$$\text{Mass Percentage} = \frac{\text{Mass of Solute}}{\text{Mass of Solution}} \text{ X } 100$$

**Empirical Formula** - It represents the simplest whole number ratio of various atoms present in a compound. It is calculated in following steps:

1. Convert the given amount to grams
2. Calculate the number of moles for each element
3. Divide the rest of the moles with the smallest one among them
4. Write the ratio with respect to elements

**Molecular Formula** - It gives the exact number of various atoms in a given compound. It is calculated by:

1. Divide "Molar Mass" with "Empirical Formula"
2. Multiply the value with each element

**Limiting Reagent** - It is the reactant which gets consumed and thus influence the amount of product formed.

**Mole Fraction of A** = number of moles of A/(number of moles of A + number of moles of B)
**Mole Fraction of B** = number of moles of B/(number of moles of A + number of moles of B)

**Molarity:**

$$\text{Molarity (M)} = \frac{\text{No. of moles of solute}}{\text{Volume of Solution in Liters}}$$

**Molality:**

$$\text{Molality (m)} = \frac{\text{No. of moles of solute}}{\text{Mass of solvent in Kilogram}}$$

## VERY SHORT ANSWER TYPE QUESTIONS ( 1 MARKS)

1. **Suppose a length had been reported to be 31.24 cm. What is the minimum uncertainty implied in this measurement?**

Ans. The minimum uncertainty implies in this measurement is $\pm$ 0.01cm.

**2. What is the difference between 5.0g and 5.00 g?**

Ans. Though the two values seem to be equivalent but scientifically they are different 5.0g has two significant figures and hence ,its precision is 0.1 part in 5,i.e., 20 ppt .5.00 has three significant figures and hence, its precision is 0.001 parts in 5,i.e.,2 ppt.Hence,5.00g is more precise measurement than 5.0g.

**3 .How is the term material different from matter?**

Ans. Anything which has mass and occupies space is called matter. However, material corresponds to the matter which has specific use(s).

**4. If the speed of light is $3.0 \times 10^8$ $ms^{-1}$, calculate the distance covered by light in 2.00 ns.**

Ans. 2.00 ns = $2.00 \times 10^{-9}$s

Distance covered = speed x time

= $3.0 \times 10^8$ $ms^{-1}$ x $2.00 \times 10^{-9}$ s = 0.6 m.

**5. What mass of silver nitrate will react with 5.85g of sodium chloride to produce 14.35g of silver chloride and 8.5 g of sodium nitrate, if the law of conservation of mass is true?**

Ans. According to law of conservation of mass,

Mass of reactants = mass of products.

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

x g 58.5 g 14.35 g 8.5 g

x +5.85 = 14.35 + 8.5

x = 17.0 g

**6. What is the SI unit of mass? How is it defined?**

Ans. SI unit of mass is kilogram (kg). It is equal to the mass of international prototype of the kilogram.

**7. What do you mean by significant figures?**

Ans. The total number of digits in a number including the last digit whose value is uncertain is called number of significant figures.

e.g., 3.200 has 4 significant figures but 3200 has only 2 significant figures.

**8. How many significant figures should be present in the answer of the following calculations?**

$$\frac{2.5 \times 1.25 \times 3.5}{2.01}$$

Ans. Least precise term i.e., 2.5 or 3.5 has two significant figures. Hence , the answer must have two significant figures.

$$\frac{2.5 \times 1.25 \times 3.5}{2.01}$$

$= 5.4415$

$= 5.4$

**9. Why is the law of Gay Lussac's not obeyed if any reactant or product is not a gas?**

Ans. If any reactant or product is a liquid or solid , the volume occupied by them is extremely small as compared to the gas and hence, the law is not obeyed.

**10. How can you prove that red oxide of copper is not an element?**

Ans. On heating red oxide of copper in the presence of hydrogen, its mass decreases and it forms shining metallic copper which is chemically different from the original compound.

$$Cu_2O + H_2 \rightarrow 2Cu + H_2O$$

**11. What will be the mass of one atom of C- 12 in gram?**

Ans. Mass of 1 atom of $^{12}C$ = $\frac{\text{Atomic mass of C}}{\text{Avogadro's number}}$

$= \frac{12g}{6.022 \times 10^{23}}$

$= 1.9927 \times 10^{-23}$g.

**12 .Give an example of molecule in which the ratio of the molecular formula is six times the empirical formula.**

Ans. The compound is glucose. Its molecular formula is $C_6H_{12}O_6$ while empirical formula is $CH_2O$.

**13. Calculate the number of gram molecules of water in a beaker containing 576g of water.**

Ans. Molecular mass of $H_2O = 2x1+16 = 18g\ mol^{-1}$

18g of water = 1g molecule

576 g of water = $\frac{1 \times 576}{18}$

=32g molecule

**14. What is the SI unit of mole? How is the mole defined?**

Ans. The word mole (means heap or pile) was introduced by Avogadro and is defined as the amount of a substance that contains as many particles or entities as there are atoms in exactly 12g (or 0.012 kg) of the $^{12}$ C isotope

**15. Calculate the percentage of N in $NH_3$ molecule.**

Ans. Molar mass of $NH_3 = 14+1x3 = 17g\ mol^{-1}$

Percentage of N = $\frac{\text{mass of N in } NH_3 \times 100}{\text{molar mass of } NH_3}$

$= \frac{14x100}{17}$

$= 82.35\%$

**16. Why are the atomic masses of most of the elements fractional?**

Ans. It is because most of the elements occur in nature as a mixture of isotopes and their atomic masses are the average relative atomic masses of the isotopes depending on their abundance.

**17. Calculate molecular mass of glucose ($C_6H_{12}O6$) molecule.**

**Or**

**A black dot used as a full stop at end of a sentence has a mass of about one attogram. Assuming that the dot is made up of carbon, calculate the approximate number of carbon atoms present in the dot?**

Ans. Molecular mass of glucose ($C_6H_{12}O_6$) = 6 (12.011 u) + 12 (1.008 u) + 6 (16.00 u)

= (72.066 u) + (12.096) + (96.00 u)

= 180.162 u

OR

Mass of carbon in the dot = 1 attogram = $10^{-18}$

Gram atomic mass of carbon = 12 g, i.e., 12 g of carbon contains $6.022 \times 10^{23}$ atoms of carbon.

$10^{-18}$ g of carbon will contain atoms = $\frac{6.022 \times 10^{23} \times 100}{12}$

$= 5.02 \times 10^4$ atoms.

**18. In the combustion of methane, what is the limiting reactant and why?**

Ans. Methane is the limiting reactant because the other reactant is oxygen of the air which is always present in excess. Thus, the amounts of $CO_2$ and $H_2O$ formed will depend upon the amount of $CH_4$ burnt.

**19. Hydrogen gas is prepared in the laboratory by reacting dilute HCl with granulated zinc. Following reaction takes place.**

**$Zn + 2HCl \rightarrow ZnCl_2 + H_2$**

**Calculate the volume of hydrogen gas liberated at STP when 32.65 g of zinc reacts with HCl. 1 mole of a gas occupies 22.7 L volume at STP ; atomic mass of Zn = 65.3 u.**

Ans. From the equation, 65.3 g of zinc liberates 22.7 L of hydrogen.

So, 32.65 g of zinc will liberates $H_2$ = 32.65 g x $\frac{22.7 \text{ L}}{2}$

= 11.35 L

**20. Volume of a solution changes with change in temperature, then, will the molality of the solution be affected by temperature? Give reason for your answer.**

Ans. No, molality of solution does not change with temperature since mass remains unaffected with temperature.

**21. What do you understand by stoichiometric coefficients in a chemical equation?**

Ans. The coefficients of reactants and products involved in a chemical equation represented by the balanced form, are known as stoichiometric coefficients.

e.g., $N_{2\,(g)} + 3\ H_{2(g)} \rightarrow 2\ NH_{3(g)}$

**22. If 2L of $N_2$ is mixed with 2 L of $H_2$ at a constant temperature and pressure, then what will be the volume of $NH_3$ formed?**

Ans. $N_{2(g)} + 3\ H_{2(g)} \rightarrow 2\ NH_{3(g)}$

1 L of $N_2$ reacts with 3 L of $H_2$.

2 L of $N_2$ will react with 6L of $H_2$ butwe have only 2 L of $H_2$ , therefore ,$H_2$ isthe limiting reagent.

3L of $H_2$ gives 2 L of $NH_3$.

2L of $H_2$ gives = $\frac{2 \times 2}{3}$

$= \frac{4}{3}$

= 1.33 L of $NH_3$

**23. How many moles of iron can be made from $Fe_2O_3$ by the use of 16 moles of carbon monoxide in the following reaction?**

**$Fe_2O_3 = 3CO \rightarrow 2\ Fe + 3\ CO_2$**

Ans. $Fe_2O_3 + 3CO \rightarrow 2\ Fe + 3CO_2$

16 mol ?

3 moles of CO are used to make 2 mol of Fe.

16 moles of CO are used to make $\frac{2 \times 16}{3}$

= 10.67 mol.

**24. What is the difference between molality and molarity?**

Ans. MOLARITY: It is defined as the number of moles of the solute in 1 L of the solution. Thus,

MOLARITY (M) = $\frac{\text{number of moles of solute}}{\text{Volume of solution in litres}}$

MOLALITY: It is defined as the number of moles of solute present in 1 kg of solvent. It is denoted by m.Thus,

MOLALITY (m) = $\frac{\text{number of moles of solute}}{\text{Mass of solvent in 1 kg}}$

## Short answer type questions (2 marks)

**Q1.Match the followings with their multiples.**

| s.no | Prefix | Multiples |
|---|---|---|
| I | Micro | $10^6$ |
| II | Deca | $10^9$ |
| III | Mega | $10^{-6}$ |
| IV | Giga | $10^{-15}$ |
| V | femto | 10 |

Ans. Micro = $10^{-6}$, deca = 10, mega = $10^6$, giga = $10^9$, femto = $10^{-15}$.

**Q2. Round up the following up to three significant figures.**

i. 34.216

ii. 10.4107

iii. 0.04597
iv. 2808

Ans.

i. 34.2
ii. 10.4
iii. 0.0460
iv. 2810

**Q3. Pressure is determined as force per unit area of the surface .the SI unit of pressure, Pascal is as shown bellow.**

$1Pa = 1Nm^{-2}$.

If the mass of air at sea level is 1034 gcm-2, calculate the pressure in Pascal.

Ans. Pressure is the force or weight per unit area.

$$\text{Pressure} = (1034g \times 9.8ms - 2)/cm2 = \frac{1034kg \times 100 \times 100 \times 9.8ms-2}{1000m2}$$

$= 101332.0Nm^{-2}$

= 1.01332 x 105 Pa

**Q4. Express each of the following in S.I. units**

**i. 14 pound per square inch**
**ii. 100 mile per hour**
**iii. 5 feet 2 inch**
**iv. 150 pound**

Ans.

i. 1 pound per square inch $= 6894.76Nm^{-2}$
.'. 14 pound per square inch=14x6894.76

$=96526.64Nm^{-2}$

ii. 1 mile/h=$\frac{1.6*10^3}{3600s}$
.'. 100 miles/h =44.7 m/s.

iii. 5 feet 2 inch = 1.5776m.
iv. 1 pound=0.454 kg
.'. 150 pounds = 68.1 kg

**Q5. The water level in a metric measuring cup is 0.75 L before the addition of pebble weighing 150 gm .the water level after submerging the pebble is 0.82 L .determine the density of pebble.**

Ans. The volume displaced by the pebble = 0.82 -0.75

= 0.07L =70 mL.

Mass of pebble = 150 g

Therefore density of the pebble is

$$\text{Density} = \frac{mass}{volume}$$

$$= \frac{150}{70} = 2.14 gmL^{-1}$$

**Q6. 1.375 g of cupric oxide was reduce by heating in a current opf hydrogen and the weight of copper that remained was 1.098 g .in another experiment, 1.179 g of cu was dissolved in nitric acid and the resulting copper nitrate was converted into cupric oxide by ignition. The weight of cupric oxide formed was 1.476g.show that these results illustrate the law of constant composition.**

Ans. In the first experiment, $CuO + H_2O \rightarrow Cu + H_2O$

1.375g 1.098g

Since 1.375g CuO is reduced to 1.098g Cu , hence, the mass of oxygen is

1.375-1.098=0.277g

Thus, the percentage of Cu = $\frac{1.098}{1.375}$ x 100 = 79.85%

In the second experiment, mass of CuO = 1.476 g

Mass of Cu = 1.179 g

.'. Mass of oxygen = 1.476 – 1.179 0.297 g

Thus, the percentage of Cu = 1.179/1.476 x 100 = 79.87%

'.'% of Cu is same in both the experiments, hence the law of constant composition is proved.

**Q7. If ten volume of dihydrogen gas react with five volume of dioxygen gas, how many volume of water vapour would be produced?**

Ans. $2H_2 (g) + O2 \rightarrow 2H_2O (g)$

According to Gay Lussac's law of gaseous volumes, 2 volume of dihydrogen react with 1 volume of $O_2$ to produce 2 volumes of water vapor.

.'. 10 volumes of H2 on reaction with 5 volume of O2 will produce 10 volume of water vapour.

**O8. Air contains 20% of oxygen by volume. Calculate the theoretical volume of air which will be required for burning completely 500 m3 of acetylene gas. All volumes are measured under the same combination of temperature and pressure.**

**Ans.**

| | | | |
|---|---|---|---|
| **$2C_2H_2$** | **$+5O_2$** | **$\rightarrow 4CO_2$** | **$+2H_2O(g)$** |

| | | | |
|---|---|---|---|
| 2 vol.<br><br>Or 1 vol.<br>50 500 $m^3$<br>500 $m^3$ | 5 vol.<br><br>5/2 vol.<br>5/2 × 500 $m^3$<br>1250 $m^3$ | 4 vol.<br><br>4/2 vol.<br>4/2 × 500 $m^3$<br>1000 $m^3$ | 2 vol.<br>(Gay Lussac's law)<br>2/2 vol.<br>2/2×500 $m^3$ |

Thus, 1250 $m^3$ oxygen is required for burning 500 m3 of acetylene. But the % of $O_2$ in air is 20%.

.'. Vol. of air required = 1250 ×100/20 = 6250 $m^3$.

**Q9. 5.975 g of the higher oxides of metal gave 5.575g of lower oxide on heating .the quantity of the lower oxide gave 5.175 g of metal on reduction . prove that these results are in accordance with the law of multiple proportion .**

**Ans.**As 5.575 g of lower oxide on reducing gives 5.175 g of the metal, the mass of the oxygen is 5.575 – 5.175 = 0.4 g.

For 1g of metal, mass of oxygen is 0.4 / 5.175 = 0.077 g

In case of higher oxides, mass of metal is 5.175 g

Mass of oxygen is 5.975 – 5.175 = 0.8 g

For 1 g of metal mass of oxygen is 0.8/ 5.175 = 0.155 g

For a given mass of metal the ratio of oxygen is 0.777: 0.155 or 1:2. Hence the law multiple proportions are proved.

**Q10. Which one of the following will have the largest no. of atoms?**

i. **1 g Au (s)**
ii. **1g Na(s)**
iii. **1gLi (s)**
iv. **1g of $Cl_2$ (g )**

**Ans.**

i. **1 g Au = 1/ 197 mole atoms of Au = 1/197 × 6.022** × $10^{23}$ atoms of Au .
ii. **1 g Na = 1/ 23 mole atoms of Na = 1/23× 6.022** × $10^{23}$ atoms of Na.
iii. **1 g of Li = 1/7 mole of Li = 1/7** ×6.022 × $10^{23}$ atoms of Li.
iv. **1 g of $Cl_2$ = 1/71 mole molecules of $Cl_2$= 1/71** ×6.022 × $10^{23}$ atoms of $Cl_2$.

**Q11. What is the molecular mass of a substance each molecule of which contains 9 atoms of carbon, 13 atoms of hydrogen and 2.33** × $10^{-23}$ g other component ?

Ans. Mass of 9 atoms of carbon = 9 × 12 amu = 108 u.

Mass of 13 atoms of hydrogen = 13 x 1 amu = 13 u.

Mass of 2.33 × $10^{-23}$ g other component = ( 1u) × ($\frac{2.33\times10^{-23}\ g}{1.66\times10^{-24}\ g}$).

= 14. 04 u

Molecular mass of the substance= ( 108+13+14.04)u = 135.04 u.

**Q12. Calculate the atomic mass of chlorine using the following data :-**

| **Isotope** | **%natural abundance** | **Molecular mass** |
|---|---|---|
| **$Cl^{35}$** | **75.77** | **34.9689** |
| **$Cl^{37}$** | **24.23** | **36.9659** |

Ans. '.' Average atomic mass,

$$\bar{A} = \sum fi \, . \, Ai = f_1 \, . \, A_1 + f_2 \, . \, A_2 + \ldots\ldots$$

Avetrage atomic mass,

$$\bar{A} = 0.7577 \times 34.9689 + 0.2423 \times 36.9659$$

$$= 26.4959 + 8.9568 = 35.4527 \, .$$

**Q13 calculate the no. of moles in the following masses of :-**

i. **1.46 metric ton of Al**
ii. **7.9 mg of Ca**

Ans.

i. 1.46 metric ton of Al = $1.46 \times 10^3 \times 10^3$ g of Al

$= 1.46 \times 10^6$ g

Atomic mass of Al = 27

Moples of Al = $\frac{mass\ of\ Al}{atomic\ mass} = \frac{1.46 \times 10^6}{27}$

$= 5.41 \times 10^4$ mol.

ii. 7.9 mg of Ca = 7.9 x 10-3 g of Ca
Atomic mass of Ca = 40.1

Moles of Ca= $+\text{-}\frac{mass\ of\ Ca}{atomic\ mass} = \frac{7.9 \times 10^{-3}}{40.1}$

$= 1.97 \times 10^{-4}$ mol.

**Q14. Chlorophyll the green coloring matter of plants contains 2.68 % magnesium by weight. Calculate the no. of Mg atoms in 2g of chlorophyll.**

**Ans.** Mass of chlorophyll = 2g .

% of Mg = 2.68 g .

Mass of Mg in 2g of chlorophyll = $\frac{2.68 \times 2}{100}$ = 0.054 g

$6.022 \times 10^{23}$ atoms of Mg = 24 g

.'. 24 g of Mg contains $6.022 \times 10^{23}$ atoms.

.'. 0.054 g of Mg contains = $\frac{6.022 \times 1023}{24} \times 0.054$

$= 1.3 \times 10^{23}$ atoms.

**Q15. A compound on analysis was found to contain C = 34.6 % , H = 3.85 % , and O = 61.55% . calculate its empirical formula.**

Ans,. Calculation of the empirical formula :-

| Element | Percentage | Atomic mass | Gram atoms | Molar ratio | Simplest whole no. ratio |
|---|---|---|---|---|---|
| C | 34.6 | 12 | 34.6/ 12= 2.88 | 2.88/2.88=1 | 3 |
| H | 3.85 | 1 | 3.85 | 3.85/2.88= 1.335 | 4 |
| O | 61.55 | 16 | 61.55/16=3.85 | 3.85/2.88= 1.335 | 4 |

.'. the simplest whole no. ratios of the different elements are : C:H:O :: 3:4:4 and the empirical formula of the compound = $C_3H_4O_4$

**Q16. A flask P contain 0.5 moles of $O_2$ , another flask Q contains 0.4 moles of O3. Which of the two flask contain greater no. of $O_2$ atoms?**

Ans. 1 molecule of $O_2$ = 2 atoms of $O_2$

1 molecule of $O_3$ = 3 atoms ofg $O_2$

In flask P , 1 mole of $O_2$ = $6.022 \times 10^{23}$ molecules

.;. 0.5 mol of $O_2$= $6.022 \times 10^{23} \times 0.5$ molecules = $6.022 \times 10^{23} \times 0.5 \times 2$ atoms

$= 6.022 \times 10^{23}$ atoms

In flask Q , 1 mole of $O_3$ = $6.022 \times 10^{23}$ molecules

0.4 mole of $O_3$ $6.022 \times 10^{23} \times 0.4$ molecules = $6.022 \times 10^{23} \times 0.4 \times 3$ atoms

$= 7.23 \times 10^{23}$ atoms

.;. Flask Q has greater no. of $O_2$.

**Q17. What is the concentration of sugar in mole $L^{-1}$ if its 20 g are dissolved in water to make a final volume up to 2L?**

Ans. Molar mass of swgar, C12H22O11

, m= (12 X 1 2.01) + (22 X 1.0079) + (11 X 16 )

= 342.294 g $mol^{-1}$ = 342

Given, w = 20 g, V = 2L

$$\text{Molarity} = \frac{w}{m \times V\ (l)} = \frac{20}{342 \times 2}$$

$$= 0.0292 \text{ mol } L^{-1}$$

$$= 0.0292 \text{ M}$$

**Q18. How much can be obtained from 100 g of CuSO4?**

Ans. Molar mass of $CuSO_4$ = 63.54 + 32.06+ (4X16)

$$= 159.6 \text{g mol}^{-1}$$

159.6g $CuSO_4$ contains 63.54g Cu

1g $CuSO_4$ will contain 63.54 / 159. 6 g Cu

.'. 100 g $CuSO_4$ will contain 63.54 X 100 / 159.6 = 39.81 g Cu

**Q19. How are 0.50 mol $Na_2CO_3$ and 0.50 M $Na_2CO_3$ different?**

Ans. Molar mass of $Na_3CO_3$ = ( 2 X 22.99) +12.01+03 X 16

$$=105.99 \text{g mol}^{-1}$$

0.50 mol $Na_2CO_3$ = 0.50 X 106 = 53 g $Na_2CO_3$

0.50 M $Na_2CO_3$ means 53g $NA_2\ CO_3$ is present in 1L of Solution.

**Q20. How many gram of Na2CO3 should be dissolved to make 100 cm3 of 0.15 M $Na_2CO_3$solution ?**

Ans. 1000 cm3 of 0.15 M Na2CO3 contains = 0.15 mol

100 cm3 of 0.15 M Na2CO3 will contain $NA_2CO_3$ = 0.15 /1000 X 100 = 0.015 mol

Mass of $Na_2CO_3$ = 0.015 X 106 = 1.59 g.

## Short Answer Type Question (3Marks)

1. **Express the following in scientific notation.**
   (i) 0.0048 (ii) 234000 (iii) 8008 (iv) 5000 (iv) 6.0012 (v) 6.0012 (vi) 3
   - To convert digit into scientific notation (N X $10^n$) shift the decimal points just after a non-zero digit.
   - If shifting is towards right, n has negative value and if shifting is towards left, n has positive value.

Sol.

(i) $4.8 * 10^{-3}$ (ii) $2.34 * 10^5$ (iii) $8.008 * 10^3$ (iv) $5 * 10^3$ (v) $6.0012 * 10^0$ (vi) $3 * 10^0$

2. **The mass of precious stones is expressed in items of "carat". Given that 1 carat = 3.168 grain and 1g=15.4 grain, calculate the total mass of a ring in gram and kilogram which contains 0.500carat diamond and 7.00 g gold.**

Sol. Mass of diamond in ring ,

$m_1$=0.500 carat

Mass of diamond in gramwill be

$$=0.500\text{carat} * \frac{3.168\text{grains}}{1\text{carat}} * \frac{1\text{g}}{15.4\text{ grains}}$$

$$= \frac{(0.500 * 3.168)}{1.54}$$

=0.10g

Mass of gold in ring , $m_2$= 7.00g

The total mass of the ring

$= m_1+m_2 = 0.10\text{ g} + 7.00\text{ g}$

=7.10 g

$= 7.10\text{ g} * \frac{1\text{ kg}}{1000\text{ g}}$

= 0.0071 kg

3. **Perform the following calculation and express the result s to proper number or significant figures.**

(i) $144.3\ m^2 + (2.54\ m * 8.4\ m)$

(ii) $(4.05 * 10^2 mL) - (0.0225 * 10^2 mL)$

(iii) $(3.50 * 10^2\ cm)\ (4.00 * 10^6\ cm)$

Sol. (i) $144.3\ m^2 + (2.54\ m * 8.4\ m)$

2.54m * 8.4m = 21.336 0r 21 m

(up to 2 significant figure)

$144.3\ m^2$

$\underline{21\ m^2}$

$165.3\ m^2$

(As 144.3 contains (1) one digit (SF) after the decimal points.)

(ii) $(4.05 * 10^2 mL) - (0.0225 * 10^2 mL)$

$= 4.0275 * 10^2 mL) - (0.0225 * 10^2 mL)$

( up to decimal place as in 4.05)

(iii) $(3.50 * 10^2\ cm)\ (4.00 * 10^6\ cm)$

$= 14.0 * 10^8\ cm^2$ (up to 3 significant figure)

4. **Convert The following into basic units.**

**(i) 28.7 pm (ii) 15.15 μs (iii) 25365 mg**

[NCERT Exemplar]

The basic unitsfor length is meter (m), for time in second (s) and for mass is kilogram (kg).

Sol.

(i) $28.7pm * \frac{10^{-12}\ m}{1pm} = 2.87 * 10^{-11} m$

(ii) $15.15\mu s * \frac{10^{-6} s}{1\mu s} = 1.515 * 10^{-5} s$

(iii) $25365\ mg * \frac{1g}{1000mg} * \frac{1kg}{1000g} = 2.5365 * 10^{-2}\ kg$

(1x3)

**5. If two elements can combine to from more than one compound,the masses of one elements that combine with a fixed mass of the other element, are in whole number ratio.**
**1. Is this statement true? 2.If yes, according to which law?**

**3. Give one example related to this law.**

Sol. (i)Yes

(ii) According to the law of multriple proportions.

iii. $H_2 + O_2 \text{---------} > H_2O$

2g 16g 18g

$H_2 + O_2$ ----------$H_2O_2$

2g 32g 34g

Here,masses of oxygen,(i.e.,16g in $H_2o$ and 32g in $H_2O_2$)which combine with fixed mass of hydrogen (2g) are in the simple ratio i.e.,16:32or1:2.

**6. Two oxides of metal contain 27.6/ 30.0/ of oxygen respectively. If the formula of the first oxide is $M_3O_4$ , find that of the second.**

Sol. In the first oxide, oxygen = 27.6

Metal = 100 – 27.6= 72.4 parts by mass.

As the formula of the oxide is $M_3O_4$ ,it means

72.4 parts by t\mass of metal = 3 atoms of metal and

4 atoms of oxygen = 27.6 parts by mass.

In the second oxide , oxygen = 30.0 parts by mass and metal = 100 -30 = 70 parts by mass.

But 72.4 parts by mass and metal = 3 atoms of metal

$\therefore$ 70 parts by mass and metal = $\frac{3}{72.4} * 70$ atoms of metal.

= 2.90 atoms of metal

Also, 27.6 parts by mass of oxygen = 4 atom of oxygen.

: 30parts by mass of oxygen = $\frac{4}{27.6}$ * 30 atom of oxygen

= 4.35 atom of oxygen

Hence, ratio of M:O in the second oxide

= 2.90 : 4.35 = 1 : 1.5 : 2: 3

:formula of the other metal oxide is $M_2O_3$

**7. 10mL of $H_2$combines with 5mL of $O_2$to from water. When 200mL of $H_2$AT STP is passed over heated CuO, the CuO loses 0.144g of its weight. Does the above data correspond to the law of constant composition?**

Sol. In the second experiment 0.144g weight is lost from CuO. This is due to the reduction of CuO into 32g oxygen occupies 22400mL volume at STP.

0.144G Oxygen will occupy=22400 * $\frac{0.144}{32}$ = 100.8m$LO_2$

It means the ratio of $H_2$ and $O_2$ in water is 200:100.8=2:1.The same ratio is in first case (10:5or2:1)Thus,the law of constant composition is proved.

**8. A box contains some identical red coloredballs,labeled as A,each weighing 2g.Another box contains identical blue coloured balls,labeled as$B_1$,each weighing 5g.Consider the combination AB,$AB_2$,$A_2B$ and $A_2B_3$ and show that law of multiple proportion is applicable.**

Sol.

| Combionation | Mass of A(g) | Mass of B(g) |
|---|---|---|
| AB | 2 | 5 |
| $AB_2$ | 2 | 10 |
| $A_2B$ | 4 | 5 |
| $A_2B_3$ | 4 | 15 |

Mass of B which is combined with fixed mass of A(say 1g)will be 2.5g,5g,1.25g and 3.75g.They are in the ratio 2:4:1:3 which is simple whole number ratio.Hence,the law of multiple proportion is applicable.

**9. Describe what you need to do in the laboratory to test(i)the law of conservation of mass,(ii)the law of definite proportion and (iii)the law of multiple proportions.**

Sol.(i)To test the law of conservation of mass,a reaction would have to be carried out in which the mass of the reactants and the mass of products are weighed and shown to be the same.

(ii)The law of definite proportions could be shown by demonstrating that no matter how a compound is obtained,the reactants remain at the same proportions by mass.This can be done by decomposing a compound and shwing that the masses of the elements present are always in the same ratio.

(iii)To test the law of multiple proportons,two different compounds made up of the same elements would have to be decomposed.If the mass of one of the elements is kept constant the masses of other elements with that of the element in different samples would have to be in small whole number ratio.

**10. 45.4L of dinitrogen reacted with 22.7L of dioxygen and 45.4L nitrous oxide was formed.The reaction is given below**

$2N_2(g)+O_2(g)$ ------->$2N_2O(g)$

Which law is being obeyed in this experiment? Write the statement of the law.

Sol. $2N_2(g) + O_2(g)$-----→$2N_2O(g)$

2V 1V 2V

$\frac{45.4}{22.7} = 2$ $\frac{22.7}{22.7} = 1$ $\frac{45.4}{22.7} = 2$

Hence the ratio between the volume of the reactants and the product in the given question is simple i.e.,2:1:2.It proves the Gay Lussac's law of gaseous volumes.

Gay Lussac's law of gaseous volumes :Refer page 9.

**Short Answer Type Question**

1. **In three moles of ethane ($C_2H_6$), calculate the following.**
   - **(i) Number of moles of carbons atoms.**
   - **(ii) Number of moles of hydrogen atoms.**
   - **(iii) No. of molecules of ethane.**

Sol. (i) 1 Mole of $C_2H_6$ contains 2 moles of carbon atoms.

∴ Number of moles of carbon atoms in 3 moles of

$C_2H_6$=2*3 = 6

(ii) 1 mole of $C_2H_6$ contains 6 moles of hydrogen atoms.

∴ Number of moles of Hydrogen atoms in 3 moles of

$C_2H_6$=3*6 = 18

(iii). 1 mole of $C_2H_6$ contains $6.022*10^{23}$ molecules of ethane.

∴ Number of ethane molecules in 3 mole of

$C_2H_6=3*6.023*10^{23}$

$= 1.8069*10^{24}$

**2. Calculate the molecular mass of the following.**

**(i) $H_2O$ (ii) $CO_2$ (iii) $CH_4$**

Sol. (i) Molecular mass of $H_2O$= 2 * atomic mass of hydrogen +1 * atomic mass of oxygen

= 2*1.0079 u + 16.00 u = 18.0158 u

(ii) Molecular mass of $CO_2$ =1*atomic mass of Carbon+2*atomic mass of oxygen

= 1*12.01 u + 2*16.00 u = 44.01 u

(iii) Molecular mass of CH4 = 1*atomic mass of Carbon+4*atomic mass of Hydrogen

= 1* 12.01 u + 4*1.0079 u

= 16.0416

**3. Determine the molecular formula of an oxide of iron in which the mass per cent of iron and oxygen are 69.9 and 30.1 respectively, Given that molar mass of the oxide is 159.898 $mol^{-1}$.**

Sol.

| Element | Symbol | % by mass | Atomic mass | Relative number of moles of element | Simple molar ratio | Simple whole number molar ratio |
|---|---|---|---|---|---|---|
| Iron | Fe | 69.9 | 55.85 | $\frac{69.9}{55.85} = 1.25$ | $\frac{1.25}{1.25} = 1$ | 1*2 = 2 |
| Oxygen | O | 30.1 | 16.00 | $\frac{30.1}{16.00} = 1.88$ | $\frac{1.88}{1.25} = 1.5$ | 1.5*2 = 3 |

$\therefore$ Empirical Formula = $Fe_2O_3$ .

Empirical formula mass of $Fe_2O_3$ = (2*55.85)+ (3*16.00)= 159.7 g $mol^{-1}$

$$n = \frac{\text{molar mass}}{\text{empirical formula mass}} = \frac{159.8}{159.7} = 1$$

Hence,molecular formula is same as empirical formula i.e., $Fe_2O_3$ .

**4. Calculate3 the total number of electronic present in 1.6 of methane.**

Sol. (i) Molar mass of methane ($CH_4$) =12+4*1 = 16g

16g of methane contains molecules = $6.022*10^{23}$

1.6 g of methane will contain molecules

$$= \frac{6.022*10^{23}}{(16\ g)} * (1.6\ g)$$

$$= 6.022*10^{22}$$

(ii). Number of electrons in $6.022*10^{22}$ molecules of methane.

1 molecule of methane contains electrons = 6+4 = 10

$6.022*10^{22}$ molecules of methane contains electrons

$$= 6.022*10^{22} *10=6.022*10^{23}$$

**5. The average molar mass of a mixture of methane($CH_4$) and ethane ($C_2H_4$) present in the ratio of a:b is found to be 20.0 g $mol^{-1}$. If the ratio were reversed,what would be the molar mass of the mixture?**

Sol. Molar mass of $CH_4$ = 16g $mol^{-1}$

Molar mass of $C_2H_4$ = 28 g $mol^{-1}$

When they are present in the a:b,their average molar

$$\text{Mass} = \frac{a*16+b*28}{a+b} = 20 \text{ g mol}^{-1} \quad \text{(Given)}$$

i.e., 16a +28b=20(a+b)
or 4a+7b = 5(a+b)
or a=2b
or $\frac{a}{b} = \frac{2}{1} = 2:1$
if the ratio is reversed,now the ratio
a:b = 1:2
∴ Average molar mass = $\frac{1*16+2*28}{1+2}$
= 24 g $mol^{-1}$

**6. A 0.005 cm thick coating of copper is deposited on a plate of 0.5 $m^2$ total area, calculate the number of copper atoms deposited on the plate(density of copper = 7.2 g $cm^{-3}$ , atomic mass = 63.5).**

Sol. Area of plate = 0.5 $m^2$ = 0.5 * $10^4$ $cm^2$

Thickness of coating = 0.005 cm

Volume of copper deposited = 0.5 * $10^4$ * 0.005 = 25 $cm^3$

Mass of copper deposited = 25* 7.2 = 180 g

Now,63.5 g of copper will contain atoms = $6.022*10^{23}$

∴ 180 g of copper will contain atoms = $\frac{6.022*10^{23}}{63.5} * 180 = 1.71* 10^{24}$ Atoms

7. **A Crystalline saltwhen heated become anhydrous and losses 51.2% of its weight. The anhydrous salt on analysis gave the percentage composition as Mg = 20.0%, S= 26.66% and O=53.33%.**

Sol.

| Element | % of mass | Atomic mass | Relative no. of moles element | Simple molar ratio |
|---|---|---|---|---|
| Mg | 20 | 24 | $\frac{20}{24} = 0.8333$ | 1 |
| S | 26 | 32 | $\frac{26}{32} = 0.8125$ | 1 |
| O | 53.33 | 16 | $\frac{53.33}{} = 3.333125$ | 4 |

The empirical formula of the anhydrous salt comes out to be $MgSO_4$ Empirical formula mass = 120 Molecular mass = 120.

Hence,molecular formula = $MgSO_4$

As crystalline salt on becoming anhydrous loses 51.2% by mass,this means 48.8 g of anhydrous salt contains $H_2O$ = 51.2 g. Therefore, 120 g of anhydrous salt contains

$= \frac{51.2}{48.8} * 120\ g = 126\ g = \frac{126}{18}$ molecules = $7H_2O$ molecules

Hence,molecular formula of crystalline salt = $Mgso_4 . 7H_20$

SHORT ANSWER TYPE QUESTIONS [3 Marks]

1. **Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation, $N_2(g) + 3H_2(g) \text{---------} > 2NH_3(g)$.**

   **(i) Calculate the mass of ammonia produced if $2.00*10^3$g dinitrogen reacts with $1.00*10^3$ g of dihydrogen.**

   **(ii) Will any of the two reactants remain unreacted?**

   **(iii) If yes, which one and what would be its mass ?**

Sol. $N_2(g) + 3H_2(G) \text{------} > 2NH_3(G)$

1 mol 3 mol 2 mol

28g 6g 34g

$28gN_2$ reacts with $6gH_2$

∴ 1 g $N_2$ will react with $\frac{6}{28}$ g$H_2$

∴ 2000 g$N_2$ will react with $\frac{2000*6}{28}$ = 428.57 g$H_2$

Hence,$N_2$is the limiting reagent and H2 is in excess,$N_2$ limits the amount of ammonia produced.

28g $N_2$ produces 34g $NH_3$

and 1g $N_2$ produces $\frac{34}{28}$ g $NH_3$

∴ 2000g$N_2$ will produce $\frac{34}{28}$ * 2000 = 2428.57 g $NH_3$

(ii) $H_2$ is in excess so it will remain unreacted.

iii. Amount of $H_2$ remain unreacted

= 1000 – 428.57 = 571.43 g

**2. Calculate the amount of carbon dioxide that could be produced when.**

**(i) 1 Mole of carbon is burnt in air.**

**(ii) 1 Mole of carbon in burnt in 16 g of dioxygen.**

**(iii) 2 Moles of carbon are burnt in 16 g of dioxygen.**

- **In order to find the moles of different elements/compounds taking part in the reaction, write a balanced chemical equation for combustion of carbon in dioxygen(air).**
- **Calculate the amount of $CO_2$ Produced with the help of information available from the balanced chemical equation.**

Sol. $C(s) + 0_2(g)$ --------> $CO_2(g)$

1 mol 1 mol 1 mol

32 g 44 g

(i) According to the equation when 1 mole of carbon is burnt completely,$CO_2$Produced is 44 g.

(ii) 1 Mole of Carbon requires 32 dioxygen from the reaction,But we have only 16g dioxygen,Hence, dioxygen is the limiting reagent, so the amount of $CO_2$ Produced by 16g dioxygen is 22 g.

(iii) In this case again, dioxygen is the limiting reagent. 16 g dioxygen can react only with 0.5 mole of carbon and produce 22g$CO_2$.

**3. Commercially available concentrated hydrochloric acid contains 38% HCL by mass.**

**(i). What is molarity of the solution(density of solution = 1.19 g $mL^{-1}$)?**

**(ii). What Volume of concentrated HCL is requied to make 1.0 L of an 0.10 M HCL?**

Sol. (i) 38% HCL by mass means 38 g of HCL is present in 100 g of solution.

Volume of Solution $= \frac{\text{mass}}{\text{density}} = \frac{100}{1.19} = 84.03$ mL.

Moles of HCL $= \frac{38}{36.5} = 1.04$

Molarity $= \frac{1.04 * 1000}{84.03} = 12.38$ M

(ii) From the molarity equation, $M_1 V_1 = M_2 V_2$

($M_1 V_1$: Acid1; $M_2 V_2$: Acid 2)

$12.38 \text{ M} * V_1 = 0.10 \text{ M} * 1.0 \text{ L}$

$\therefore \quad V1 = \frac{0.1 * 1.0}{12.38} = 0.00808 \text{ L} = 8.08 \text{ cm}^3$

**4. If the density of methanol is 0.793 kg $L^{-1}$ , What is its volume needed for making 2.5 L of its 0.25 M solution?**

Sol. Given, $d = 0.793 \text{ kg L}^{-1} = 0.793 * 10^3 \text{ gL}^{-1}$

Final Volume, $V_2 = 2.5$ L

Final Molarity, $M_2 = 0.25$ M

Molarity of Initial Solution $M_1 = ?$

Initial volume $V_1 = ?$

Molar mass of methanol,

CH3OH $= (1 * 12.01) + (4 * 1.0079) + 16.00$

$= 32.0416 = 32 \text{ g mol}^{-1}$

Molarity $= \frac{0.793 * 10^3 \text{ gL}^{-1}}{32 \text{ g mol}^{-1}} = 24.781 \text{ mol L}^{-1}$

$M_1 V_1 = M_2 V_2$

$24.781 * V_1 = 0.25 * 2.5$

$V_1 = \frac{0.25 * 2.5}{24.781} = 0.02522$ L

$= 25.22$ mL

**5. Chlorine is prepared in the laboratory by treating manganese dioxide (MnO2) with aqueous hydrochloric acid according to the reaction,**

**$4HCL(aq) + MnO_2(s) \text{ -------> } 2H_2O(l) + MnCl_2(aq) + Cl_2(g)$**

**How many gram of HCL reacts with 5.0 g of manganese dioxide?**

Sol. $4HCL(aq) + MnO_2(s) \text{ ----------> } 2H_20(l) + MnCl_2(aq) + Cl_2(g)$

4 * 36.5 87 g

According to the balanced chemical equation,

87 g of $MnO_2$ reacts with 4 * 36.5 g HCL

5 g of $MnO_2$ will react with $\frac{4 * 36.5 * 5}{87}$ = 8.39 g HCL

**Short Answer Type Question (3Marks)**

5. **Convert The following into basic units.**
   **(i) 28.7 pm (ii) 15.15 µs (iii) 25365 mg**

   **The basic unitsfor length is meter (m), for time in second (s) and for mass is kilogram (kg).**

   Sol.

   (iv) $28.7\text{pm} \times \frac{10^{-12}\text{ m}}{1\text{pm}} = 2.87 \times 10^{-11}\text{m}$

   (v) $15.15\mu\text{s} \times \frac{10^{-6}\text{s}}{1\mu\text{s}} = 1.515 \times 10^{-5}\text{ s}$

   (vi) $25365\text{ mg} \times \frac{1\text{g}}{1000\text{mg}} \times \frac{1\text{kg}}{1000\text{g}} = 2.5365 \times 10^{-2}\text{ kg}$ (1x3)

6. **Express the following in scientific notation.**
   **(iii) 0.0048 (ii) 234000 (iii) 8008 (iv) 5000 (iv) 6.0012 (v) 6.0012 (vi) 3**
   - **To convert digit into scientific notation (N X $10^n$) shift the decimal points just after a non-zero digit.**
   - **If shifting is towardsright, n has negative value and if shifting is towards left, n has positive value.**

Sol.

(i) $4.8 \times 10^{-3}$ (ii) $2.34 \times 10^{5}$ (iii) $8.008 \times 10^{3}$ (iv) $5 \times 10^{3}$ (v) $6.0012 \times 10^{0}$ (vi) $3 \times 10^{0}$

7. **The mass of precious stones is expressed in items of "carat". Given that 1 carat = 3.168 grain and 1g=15.4 grain, calculate the total mass of a ring in gram and kilogram which contains 0.500carat diamond and 7.00 g gold.**

Sol. Mass of diamond in ring ,

$m_1$=0.500 carat

Mass of diamond in gramwill be

$$=0.500\text{carat} \times \frac{3.168\text{grains}}{1\text{carat}} \times \frac{1\text{g}}{15.4\text{ grains}}$$

$$=\frac{0.500 \times 3.168}{1.54}$$

$$=0.10\text{g}$$

Mass of gold in ring , $m_2$= 7.00g

The total mass of the ring

$= m_1+m_2 = 0.10\text{ g} + 7.00\text{ g}$

$=7.10$ g

= 7.10 g X $\frac{1\ kg}{1000\ g}$

= 0.0071 kg

**8. Perform the following calculation and express the result s to proper number or significant figures.**

**(iv) 144.3 $m^2$ + (2.54 m X 8.4 m )**

**(v) (4.05 X $10^2$mL ) – (0.0225 X$10^2$mL )**

**(vi) (3.50 X $10^2$ cm) (4.00 X $10^6$ cm)**

Sol. (i) 144.3 $m^2$ + (2.54 m X 8.4 m )

2.54m X 8.4m = 21.336 0r 21 m

(up to 2 significant figure)

144.3 $m^2$

21 $m^2$

165.3 $m^2$

(As 144.3 contains (1) one digit (SF) after the decimal points.)

(iv) (4.05 X $10^2$mL ) – (0.0225 X$10^2$mL )

= 4.0275 X $10^2$mL ) – (0.0225 X $10^2$mL)

( up to decimal place as in 4.05)

(v) (3.50 X $10^2$ cm) (4.00 X $10^6$ cm)

= 14.0 X $10^8$ $cm^2$ (up to 3 significant figure)

**9. Two oxides of metal contain 27.6/ 30.0/ of oxygen respectively. If the formula of the first oxide is $M_3O_4$ , find that of the second.**

Sol. In the first oxide, oxygen = 27.6

Metal = 100 – 27.6= 72.4 parts by mass.

As the formula of the oxide is $M_3O_4$ ,it means

72.4 parts by t\mass of metal = 3 atoms of metal and

4 atoms of oxygen = 27.6 parts by mass.

In the second oxide , oxygen = 30.0 parts by mass and metal = 100 -30 = 70 parts by mass.

But 72.4 parts by mass and metal = 3 atoms of metal

: 70 parts by mass and metal = $\frac{3}{72.4}$ X 70 atoms of metal.

= 2.90 atoms of metal

Also, 27.6 parts by mass of oxygen = 4 atom of oxygen.

: 30parts by mass of oxygen = $\frac{4}{27.6}$ X30 atom of oxygen

= 4.35 atom of oxygen

Hence, ratio of M:O in the second oxide

= 2.90 : 4.35 = 1 : 1.5 : 2: 3

:formula of the other metal oxide is $M_2O_3$

**10.If two elements can combine to from more than one compound,the masses of one elements that combine with a fixed mass of the other element, are in whole number ratio.**

**1. Is this statement true? 2.If yes, according to which law?**

**3. Give one example related to this law.**

Sol. (i)Yes

(ii) According to the law of multiple proportions.

iii. $H_2 + O_2$---------->$H_2O$

2g 16g 18g

$H_2 + O_2$ ----------$H_2O_2$

2g 32g 34g

Here, masses of oxygen,(i.e.,16g in $H_2o$ and 32g in $H_2O_2$)which combine with fixed mass of hydrogen (2g) are in the simple ratio i.e.,16:32or1:2.

**11.10mL of $H_2$ combine with 5mL of $O_2$to from water. When 200mL of $H_2$AT STP is passed over heated CuO, the CuO loses 0.144g of its weight. Does the above data correspond to the law of constant composition?**

Sol. In the second experiment 0.144g weight is lost from CuO. This is due to the reduction of CuO into

32g oxygen occupies 22400mL volume at STP.

0.144G Oxygen will occupy=22400 * $\frac{0.144}{}$ = 100.8m$Lo_2$
It means the ratio of $H_2$ and $O_2$ in water is 200:100.8=2:1.The same ratio is in first case (10:5or2:1)Thus, the law of constant composition is proved.

**12..A box contains some identical red colored balls, labeled as A, each weighing 2g.Another box contains identical blue coloured balls, labeled as$B_1$,each weighing 5g.Consider the combination AB,$AB_2$,$A_2B$ and $A_2B_3$ and show that law of multiple proportion is applicable.**
Sol.

| Combination | Mass of A(g) | Mass of B(g) |
|---|---|---|
| AB | 2 | 5 |
| $AB_2$ | 2 | 10 |
| $A_2B$ | 4 | 5 |
| $A_2B_3$ | 4 | 15 |

Mass of B which is combined with fixed mass of A(say 1g)will be 2.5g,5g,1.25g and 3.75g.They are in the ratio 2:4:1:3 which is simple whole number ratio. Hence, the law of multiple proportion is applicable.

9.45.4L of dinitrogen reacted with 22.7L of dioxygen and 45.4L nitrous oxide was formed. The reaction is given below

$2N_2(g)+O_2(g)$----→$2N_2O(g)$

Which law is being obeyed in this experiment? Write the statement of the law.

Sol. $2N_2(g) + O_2(g)$-----→2N2O(g)

2V 1V 2V

45.4

Hence the ratio between the volume of the reactants and the product in the given question is simple i.e.,2:1:2.It proves the Gay Lussac's law of gaseous volumes.

Gay Lussac's law of gaseous volumes :Refer page 9.

**13.Describe what you need to do in the laboratory to test(i)the law of conservation of mass,(ii)the law of definite proportion and (iii)the law of multiple proportions.**

Sol.(i)To test the law of conservation of mass,a reaction would have to be carried out in which the mass of the reactants and the mass of products are weighed and shown to be the same.

(ii)The law of definite proportions could be shown by demonstrating that no matter how a compound is obtained,the reactants remain at the same proportions by mass.This can be done by decomposing a compound and shwing that the masses of the elements present are always in the same ratio.

(iii)To test the law of multiple proportons,two different compounds made up of the same elements would have to be decomposed.If the mass of one of the elements is kept constant the masses of other elements with that of the element in different samples would have to be in small whole number ratio.

- **<u>Value Based Answer Type Question</u>**

**Q1. Shyam who live near a power plant, one day found that the fishes of his nearby river are dying. He discussed this problem with his friend who told him power plant as a cause of this problem, from which large amount of heat is released into the river and other water bodies, thus it leads to thermal pollution. Giving an example of thermal pollution he told that trout will die if the temperature of water rises about approximately $25^0$C. He satisfied from the answer of his friend and decide to discuss this problem with the people of his area and found a solution.**

**(i)What is this temperature ($25^0$C) is degree Fahrenheit?**

**(ii)What is the temperature in kelvins?**

**(iii)What values are associated with Shyam and his friends?**

Sol. (i) $\frac{℃}{5} = \frac{F-32}{9}$

We have ℃ = $25^0$C

$\therefore \quad \frac{25}{5} = \frac{F-32}{9}$ or F = $9\times 5 +32 = 45 +32 = 77^0$F

(ii) K = $^0$C +273.15 =$25^0$C +273.15 =$298.15^0$K

(iii) Shyam is a social worker , responsible , careful and his friend is intelligent ,knowledgeable , and Concerns about environmental pollution.

**Q.2 Rama's father who is a science teacher tried to teach her mole concept. For this he told her that as are 12 bananas in 1 dozen , similarly , there are $6.022\times 10^{23}$ particles in a mole which is equivalent to the molar mass of an element . He told suppose a 0.005 cm thick coating of gold is deposited on a medal of 0.5 $m^2$ total area . the density of gold is 19.30g $cm^{-3}$ and its atomic mass 196.97 .**

**(i) What is the mass of gold atoms deposited on the medal ?**

**(ii) Calculate the moles of gold deposited on the medal.**

**(iii) Calculate the number of gold atoms deposited on the medal.**

**(iv) what values are shown by Rama's father ?**

Soln. (i) Area of the medal $= 0.5\ m^2 = 0.5\times 10^4\ cm^2$

Thickness of coating $= 0.005$ cm

Volume of gold deposited $== 0.5\times 10^4 \times 0.005 = 25\ cm^3$

Mass of gold deposited $= 25 \times 19.30 = 482.5$ g.

(ii) Moles of gold deposited $\frac{\text{Mass of gold}}{\text{atomic mass}} = \frac{482.5}{196.97} = 2.45$

(iii) Number of gold atoms deposited $= 2.45 \times N_A \times 1 = 2.45 \times 6.022 \times 10^{23} \times 1$

$= 1.48 \times 10^{24}$ atoms

(iv) Rama's father is careful about her daughter's studies and teaching in playful way.

**Q.3 Ravi is Mr. Sharma's grandson. He take him regularly for routine checkup to doctor. One time doctor reported that his blood sugar level is $10 mol^{-1}$. Note that a person is declared as a diabetic if his blood**

**sugar level exceeds 160 mg/ dL when tested 2 hours after meals.**

**(i)Convert the blood sugar level of his grandfather in mg /dL units.**

**(ii)Is his grandfather diabetic?**

**(iii)What cares should be taken is case of diabeties?**

**(iv)What values are exhibited by Ravi?**

Sol . (i) Molar mass of glucose ($C_6H_{12}O_6$) $= 180 g\ mol^{-1}$

i.e $1 mol = 180$ g

$\therefore$ $1$ m mol $= 180$ mg or $1$ m mol $L^{-1} = 18$mg /dL

$\therefore$ $10$ m mol $L^{-1} = 10 \times 18$mg /dL

$= 180$ mg /dL

(ii) Since his blood sugar level is 180 mg /dL so he is diabetic .

(iii) Diabetric patient should take low calorie and low carbohydrate food. Food should be rich in fibres.

(iv) He is loving and careful about his grandfather's health.

- **<u>Long Answer Type Question</u>**

**Q1 The following data are obtained when dinitrogen and dioxygen react together to from different compounds .**

| | **Mass of dinitrogen** | **Mass of dioxygen** |
|---|---|---|
| **(i)** | **14g** | **16g** |
| **(ii)** | **14g** | **32g** |
| **(iii)** | **28g** | **32g** |
| **(iv)** | **28g** | **80g** |

**(i) Which law of chemical combination is obeyed by the above experimental data ? Give statement .**

**(ii) Fill in the blanks in the following conversion .**

**(a) 1 km =……..mm= …… pm**

**(b) 1 mg = ……..kg = …….. kg**

**(c) 1mL = …… L= …… $dm^3$**

Soln. (i)On fixing the mass of dinitrogen as 28 g, the mass of dioxygen combined are 32, 64 ,32and 80 in the four oxides . these are in the simple whole number ratio . i.e. 2 : 4 : 2 : 5.hence the given data obey the law of multiple proportions

(ii) (a) $1\text{ km} = 1\text{ km} \times \frac{1000\text{ m}}{1\text{ km}} \times \frac{100\text{cm}}{1\text{m}} \times \frac{10\text{mm}}{1\text{cm}} = 10^6\text{ mm}$

$1\text{ km} = 1\text{ km} \times \frac{1000\text{ m}}{1\text{ km}} \times \frac{1\text{pm}}{10^{-12}\text{m}} = 10^{15}\text{ pm}$

$\therefore\ 1\text{ km} = 10^6\text{ mm} = 10^{15}\text{ pm}$

(b) $1\text{ mg} = 1\text{ mg} \times \frac{1\text{g}}{1000\text{ mg}} \times \frac{1\text{kg}}{1000\text{g}} = 10^{-6}\text{ kg}$

$1\text{ mg} = 1\text{ mg} \times \frac{1\text{g}}{1000\text{ mg}} \times \frac{1\text{ng}}{10^{-9}\text{g}} = 10^{6}\text{ kg}$

$\therefore\ 1\text{ mg} = 10^{-6}\text{ kg} = 10^{6}\text{ kg}$

(c) $1\text{ mL} = 1\text{ mL} \times \frac{1\text{L}}{1000\text{ mL}} = 10^{-3}\text{ L}$

$1\text{ ml} = 1\text{ cm}^2 = 1\text{cm}^3 \times \frac{1\text{dm} \times 1\text{dm} \times 1\text{dm}}{10\text{cm} \times 10\text{cm} \times 10\text{cm}} = 10\text{ dm}^3$

$\therefore$ 1mL $= 10^{-3}$ L$= 10\text{ dm}^3$

**Q.2 A mixture of oxalic acid and formic acid is heated with concentrated $H_2SO_4$ and the gas evolved is collected. On treatment with KOH the solution the volume of solution decreases by $1/6^{th}$ .calculate the molar ratio of two acids in original mixtures.**

Sol .let x mole oxalic acid and y mole of formic acid be heated with conc. $H_2SO_4$ according to the following equations .

$$\begin{matrix}\text{COOH} \\ \text{COOH}\end{matrix} \xrightarrow{H_2SO_4/\text{ heat}} \underset{\text{x mol}}{CO(g)} + \underset{\text{y mol}}{CO_2(g)} + H_2O\ (l)$$

$$\underset{\text{xmol}}{\text{HCOOH}} \xrightarrow{H_2SO_4/\text{ heat}} CO(g) + H_2O\ (l)$$
$$\text{y mol}$$

Total moles gaseous mixture = Moles of $CO_2$= (x + y )mol + x mol = (2x = y ) mol

Now KOH absorbs only $CO_2$ i.e. x moles and the volume of the solution decreases by $1/6^{th}$ of its volume . since equal volume of gases have equal number of moles according to Avogadro law.

$\therefore \quad \frac{\text{Moles of } CO_2}{\text{Moles of both the gases}} = \frac{x}{(2x+y)} = \frac{1}{6}$

or $6x = 2x = y$

or $4x = y$

or $\frac{x}{y} = 4$

∴ Molar ratio of formic acid : oxalic acid = 4 : 1

**Long Answer Type Questions [5 Marks]**

1. **(i) The density of the water at room temperature is 0.1 g/ml. How many molecules are there in a drop of water if its volume is 0.05 mL?**
   **(ii) An alloy of iron (53.6%), nickel (45.8%) and manganese (0.6%) has a density of 8.17 g $cm^{-3}$ ,Calculate the number of Ni atoms present in the alloy of dimensions 10.0 cm * 20.0 cm * 15.0 cm.**

Sol. (i) Volume of a drop of water = 0.05 mL

Mass of a drop of water = volume * density = (0.05 mL) * (1.0 g/mL) = 0.05 g

Gram molecular mass of water ($H_20$) = 2 * 1 + 16 = 18g

18 g of water = 1 mol

∴ 0.05 g of water = $\frac{1 \text{ mol}}{(18 \text{ g})}$ * (0.05 g) = 0.0028 mol

. .1 mole of water contains molecules = 6.022 * $10^{23}$

.

0.0028 mole of water contains molecules = 6.022 * $10^{23}$ * 0.0028 = 1.68 * $10^{21}$ molecules

(ii) Volume of the alloy = (10.0 cm) * (20.0 cm) * (15.0 cm)
= 3000 $cm^3$

Mass of the alloy = density * volume = (8.17 g $cm^{-3}$) * (3000 $cm^3$) = 24510 g

Mass of Ni in the alloy = (24510 g) * $\frac{45.8}{100}$ = 11225.6 g

. . 59 g Ni have atoms = 6.022 * $10^{23}$

.

Gram atomic mass of Ni = 59g

11225.6 g of Ni have atoms = 6.022 * $10^{23}$ *(11225.6 g) = 1.15 * $10^{26}$ atoms

2. **A welding fuel gas contains carbon and hydrogen only. Burning a small sample of it in oxygen gives 3.38g carbon dioxide, 0.690 g of water and no other products. A volume 10.0 L (measured at STP) of this welding gas is found to weigh 11.6 g. Calculate**
   (i) Empirical formula (ii) Molar mass of the gas and(iii) Molecular Formula.

Sol. (i) 44g$CO_2$ = 12g carbon

3.38 g $CO_2$ = $\frac{12}{44}$ * 3.38g = 0.9218 g Carbon

18gHx0 = 2 g hydrogen

0.690g $H_20$ = $\frac{2}{18}$ * 0.690 g = 0.0767 g Hydrogen

Total mass of compound = 0.9218 + 0.0767 = 0.9985 g (because compound contains only carbon and hydrogen)

% of C in the compound = $\frac{0.9218}{}$ * 100 = 92.32

$$\% \text{ of H in the compound} = \frac{0.0767}{0.9985} * 100 = 7.68$$

Calculations of Empirical Formula

| Element | Per cent by mass | Atomic Mass | Relative Number of moles of elements | Simples molar ratio |
|---|---|---|---|---|
| C | 92.32 | 12 | $\frac{92.32}{12} = 7.69$ | $\frac{7.69}{7.68} = 1$ |
| H | 7.68 | 1 | $\frac{7.68}{1} = 7.68$ | $\frac{7.69}{7.68} = 1$ |

Hence,empirical formula = CH

(ii) Calculations for molar mass of the gas

10.0 L of given gas at STP weigh = 11.6 g

∴ 22.4 L of the given gas at STP will weigh

$$= \frac{11.6 * 22.4}{10} = 25984 \text{ g}$$

Molar mass = 25.984 = 26 g $mol^{-1}$

(iii) Empirical formula mass (CH) = 12+1 = 13

$$\therefore \quad n = \frac{\text{molecular mass}}{\text{Empirical formula mass}}$$

$$= \frac{26}{13} = 2$$

Hence, molecular formula = n * CH = 2 * CH = $c_2h_2$

**3. Arrange the following in order of their increasing masses in gram (i) One atom of silver, (ii) one gram-atom of nitrogen, (iii) one mole of calcium, (iv) one mole of oxygen molecules,(v) $10^{23}$ atoms of carbon and (vi) one gram of iron.**

Sol. (i) mole of Ag atom = 108 g = 6.022 * $10^{23}$ atoms

Mass of 6.022 *$10^{23}$ atoms of Ag = 108 g

Mass of 1 atoms of Ag = $\left[\frac{108}{6.022 * 10^{23}}\right]$

=1.793 * $10^{-22}$ g

(ii) Mass of gram atom of N = atomic mass of N in gram = 14.0 g.

(iii) Mass of a mole of Ca = atomic mass of Ca in gram = 40.0.g.

(iv) Mass of mole of oxygen molecules = molar mass of oxygen in gram = 32.0 g.

(v) Mass of mole of C atom = 12 g = 6.023 * $10^{23}$ atoms.

Mass of 6.023 * $10^{23}$atoms of C = 12g.

Mass of 1 atom of C = $\left[\frac{12}{6.023 * 10^{23}}\right]$g

Mass of 1023 atoms of $\left[\frac{108}{6.023 * 10^{23}}\right]$* $10^{23}$ = 1.992 g

(Vi) Mass of iron = 1.0 g.

Hence, the required order of increasing masses in one atom of silver < one gram of iron < 1023 atoms of C< one gram atom of nitrogen < one mole of oxygen < one mole of calcium.

**4. A compound made up of two elements A & B has A = 70% , B= 30%. Their relative number of moles in the compound are 1.25 and 1.88. Calculate**

**(i) Atomic masses of the elements A and B**

**(ii) Molecular formula of the compound, if its molecular mass is found to be 160.**

Sol. Relative number of moles of an element = $\frac{\%\text{ of the element}}{\text{Atomic mass}}$

Or atomic mass = $\frac{\%\text{ of element}}{\text{Relative number of mole}}$

∴ Atomic mass of A = $\frac{70}{1.25} = 56$

And Atomic mass of B = $\frac{30}{1.88} = 16$

Calculations of Empirical Formula

| Element | Relative number of moles | Simples molar ratio | Simplestwhole number molar ratio |
|---|---|---|---|
| A | 1.25 | $\frac{1.25}{1.25} = 1$ | 2 |
| B | 1.88 | $\frac{1.88}{1.25} = 1.5$ | 3 |

∴ Empirical formula = $A_2B_3$

Calculation of molecular formula

Empirical formula mass = 2 * 56 + 3 * 16 = 160

$n = \frac{\text{molecular mass}}{\text{empiricalforumula mass}} = \frac{160}{160} = 1$

∴ Molecular formula = $A_2B_3$

- **<u>Long Answer Type Questions [5 Marks]</u>**

1. **(i) A sample of drinking water was found to be severely contaminated with chloroform,$CHCL_3$ , supposed to be carcinogenic in nature. The level of contamination was 15 ppm(by mass).**
   **(a) Express this in per cent by mass.**
   **(b) Determine the molarity of chloroform in the water sample.**
   **(iii) Calculate the molarity of a solution of ethanol in water in which the mole fraction of ethanol is 0.040.**

   Sol. (i) (a) 15 ppm means 15 parts in one million $(10)^6$ parts.

   Therefore, % by mass = $\frac{15 * 100}{} = 1..5 * 10^{-3}\%$

(b) Molar mass of CHCl3 = 119 g $mol^{-1}$

1.5 * $10^{-3}$ % means 1.5 * $10^{-3}$ g chloroform present in 100 g sample.

Molarity, M = $\frac{w * 1000}{M * \text{volume of sample}}$

(for water, density = 1g $cm^{-3}$,so mass = volume)

$M = \frac{1.5 * 10^{-3} * 1000}{119 * 100} = 1.26 * 10^{-4}$ M

(ii) Molarity is defined as the moles of solute(ethanol) in iL of the solutions.
iL of the solutions(as it is diluted) = IL of water
Number of moles of $H_20$ = 1000g = 55.55 mol

$xH_20 = 1 - xC_2H_5OH$
$xH_20 = 1 - 0.040 = 0.96$

$XH_2O = \frac{nH_20}{nH_20 + nC_2H5OH}$

$0.96 = \frac{55.55}{55.55 + nC_2H_5OH}$
Or $nc_2h_4oh = \frac{2.222}{0.96} = 2.3145$ mol

2. **Calcium carbonates reacts with aqueous HCL to give $Cacl_2$ and $C0_2$ according to the reaction given below $Caco_3(s) + 2HCL(aq) \text{---------} > CaCl_2(aq) + CO_2(g) + H_2O(l)$**
   **What mass of $CaCl_2$ will be formed when 250 mL of 0.76 M HCL reacts with 1000 g of $CaCO_3$?**
   **Name the limiting reagent,Calculate the number of moles of $CaCl_2$ formed in the reaction.**

Sol. Number of moles of HCL = 250 mL * $\frac{0.76\ M}{1000}$ = 0.19 mol;
Number of moles of $CaCO_3$ = $\frac{1000g}{100\ g\ mol^{-1}}$ = 10mol;

For the 10 moles of $CaCO_3$ (s) number of moles of HCL
required would be 10 * $\frac{2}{1}$ = 20 mol HCL (aq)
But we have only 0.19 mole HCL(aq),hence HCL(aq) is the limiting reagent,since,2 moles HCL(aq) forms 1 mole of $Cacl_2$,therefore, 0.19 moles of HCL(aq) would give
0.19 * $\frac{1}{2}$ = 0.095 mol
Mass of $CaCl_2$ = 0.095 * 111 = 10.54 g

Que 3– **On a Particular day , the temperature recorded in New York was 60º F. What would be the equivalent temperature in Cº.**
Ans - ºC = 5/9 (ºF – 320
= 5/9 (60 – 32)
= 5 x 32/9
= 15.55
= 15.6º
Que 4- **Convert the following temperatures into degrees Fahrenheit:**
**(i) 25ºC, the room temperature (ii) 37ºC, the human body temperature.**
Ans- (i) ºF = 9/5 (ºC) + 32
=9/5(25) + 32
= 45+32
=77º
(ii) ºF = 9/5(37) + 32
= 66.6+32
= 98.6
Que5– **How many significant figures are there in each of the following numbers?**

**(i) 6.005 (ii) 8000 (iii) 0.0025**

Ans– (i) Four
(ii) Four
(iii) Two

Que 6 – **The mass of precious stones is expressed in terms of 'carat'. Given that 1 carat = 3.168 grams and 1 gram = 15.4 grains, calculate the total mass of a ring in grams and kilograms which contains 0.500 carat diamond and 7.00 gram gold.**

**Ans – The unit conversion factors to be used will be :**

1= 1 carat/3.168 grains = 3.168 grains /1 carat
1= 1 gram/ 15.4 grains = 15.4 grains/ 1 gram
0.500 carat = 0.500 x 3.168/1 x 1/ 15.4
= 0.10 gram

.
. . Total mass of the ring = 7.00+0.10 = 7.10 gram
= 7.10g x 1kg/ 1000g = 0.0071kg

**Que 7 – 90 g of KClO3 when heated produced 1.92 g of oxygen and the residue (KCl) left behind weighs 2.96 g. Show that these results illustrate the law of conservation of mass.**

Ans – Mass of KClO3 taken = 4.90g
Total mass of the products (KCl + O2) = 2.96 + 1.92 = 4.88 g
Difference between the mass of the reactant and the total mass of the products
= 4.90 – 4.88 = 0.02
This small difference may be due to the experimental error.
Thus, law of conservation of mass holds good within experimental errors.

**Que 8 – What is the difference between 5.0 g and 5.00 g ?**

Ans – Though they look to be equivalent but scientifically they are different. 5.0 g has two significant figures and hence its precision is 0.1 part in 5, i,.e., 20 p.p.t . 5.00 has three significant figures and hence its precision is 0.01 parts in 5, i.e., 2 p.p.t . Hence , 5.00 g is more precise measurement than 5.0 g.

**Que 9 – Calculate the percentage composition of the various elements in Mg SO4.**

Ans – Mol. mass of MgSO4 = 24 + 32 + 4 x 16 = 120
% of Mg = No. of parts by mass of Mg/Mol. Mass of MgSO4 x 100
= 24/120 x 100
= 20%
% of S = No. of parts by mass of S/ Mol. Mass of MgSO4 x 100
= 32/120 x 100
= 26.67%
% of O = No. of parts by mass of O/ Mol. Mass of MgSO4 x 100
= 64/120 x 100
= 53.33%

**Que 10 – Express the result of the given calculation to the appropriate number of significant figures : 3.24 x 0.08666/ 5.006**

Ans– 3.24 x 0.08666/5.006 = 0.0560883

As 3.24 has least number of significant figures, viz, 3, the result should contain 3 significant figures only.
Hence the result will be reported as 0.0561.

**Que 11 – 6.488g of lead combine directly with 1.002g of oxygen to form lead peroxide (PbO2). Lead peroxide is also produced by heating lead nitrate and it was found that the percentage of oxygen present in lead peroxide is 13.38 percent. Use these data to illustrate the law of constant composition.**

Ans – To calculate the % of oxygen in first composition
mass of peroxide formed = 6.488 + 1.002 = 7.490 g.

7.490 g of lead peroxide contain 1.002 g of oxygen

100 g of lead peroxide will contain oxygen = 1.002/7.490x 100
= 13.38g

To compare the percentage of oxygen in both the experiments.
Percentage of oxygen in $PbO_2$ in the first experiment = 13.38
Percentage of oxygen in $PbO_2$ in the second experiment = 13.38
Since the percentage composition of oxygen in both the samples of $PbO_2$ in identical, the above data illustrate the law of constant composition.

**Que 12 – Calculate the mass of iron which will be converted into its oxide ($Fe_3O_4$) by the action of 18 g of steam on it.**

Ans- The chemical equation representing the reaction is :

$3\,Fe + 4H_2O + 4H_2$

3 x 56 4 x 18

= 168g = 72g

Thus, 72 g of steam react with 168g of iron
18 g of steam will react with 168/72 x 18 = 42 g of iron
Mass of iron required = 42

**Que 13 – Naturally occurring Boron consists of two isotopes whose atomic masses are 10.01 and 11.01. The atomic mass of natural Boron is 10.81. Calculate the percentage of each isotope in natural Boron.**

Ans – Suppose the percentage of isotope with atomic mass 10.01 = X
Then percentage of isotope with atomic mass 11.01 = 100 – X

Average atomic mass = 10.01X + (100 – X) x 11.01/100
= 10.01X + 1101 – 11.01/100
= 1101 – X/ 100
1101 – X/100 = 10.81(given) or 1101 – X
= 1081 or X = 1101 – 1081 = 20
% of isotope with atomic mass 10.01 = 20%
% of isotope with atomic mass 11.01 = 80%.

**Que 14 – Chlorine has two isotopes of atomic mass units 34.97 and 36.97. The relative abundances if these two isotopes are 0.755 and 0.245 respectively. Find the average atomic mass of chlorine.**

Ans – Average atomic mass = 34.97 x 0.755 + 36.97 x 0.245
= 35.46

Que 15 – Magnetic oxide ($FeO_3$) when heated with hydrogen is reduced to iron and water is also produced. Write balanced equation for the reaction.

Ans- The reaction is : $Fe_3O_4 + H_2 \rightarrow Fe + H_2O$
The equation contains hydrogen as an elementary gas. Writing it in the atomic state, we have
$Fe_3O_4 + H \rightarrow Fe + H_2O$
The biggest formula is $Fe_3O_4$. Hence order of selection of atoms for balancing is Fe , O and H

To equalise the number of atoms of Fe and O on both sides, multiply Fe by 3 and $H_2O$ by 4.
$Fe_3O_4 + H \rightarrow 3\,Fe + 4\,H_2O$
The above equation has 8 atoms of H on the R.H.S as against 1 on L.H.S. to equalise, multiply H atoms by
8 and convert it into molecular state as $4H_2$. We have
$Fe_3O_4 + 4\,H_2 \rightarrow 3\,Fe + 4\,H_2O$
This the required balanced molecular chemical equation.

**Que 16 – Calculate the amount of water (g) produced by the combustion of 16 g of methane.**

Ans – The balanced chemical equation for the combustion of methane is:

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

1 mole = 16 g; 2 moles = 2 x 18 = 36 g

Thus, 1 mole of $CH_4$ on combustion produces 2 moles of $H_2O$.

But 1 mole of $CH_4 = 12 + 4 \times 1 = 16$ g

1 mole of $H_2O = 2 + 1 + 16 = 18$ g

2 moles of $H_2O = 2 \times 18g = 36$ g

Hence, 16 g of $CH_4$ on complete combustion produce $H_2O = 36$ g

**Que 17 – Calculate the amount of lime, Ca (OH)2, required to remove hardness of 50,000 litres of well water**
**which has been found to contain 1.62 g of calcium bicarbonate per 10 litre.**

Ans – (i) Calculation of total $Ca(HCO_3)_2$ present

10 L of water contain $Ca(HCO_3)_2 = 1.62$ g

50,000 L of water will contain $Ca(HCO_3)_2 = 1.62/10 \times 50000$ g

$= 8100$ g

(ii) Calculation of lime required. The balanced equation for the reaction involved is :

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$

1 mole; $40 + (1+12+48) = 162$ g

1 mole; $40 + (16 + 1) \times 2 = 74$ g

162 g $Ca(HCO_3)_2$ require lime = 74 g

8100 g $Ca(HCO_3)_2$ will require lime = $74/162 \times 8100$ g

$= 3700$ g

Que 18 – 3.0 g of $H_2$ react with 29.0 g of $O_2$ to form $H_2O$.

**18. (i) Which is the limiting reactant? (ii) Calculate the maximum amount of $H_2O$ that can be formed. (iii) Calculate the amount of the reactant left unreacted.**

Molecular mass of $H_2 = 2.016$

Ans - $2H_2 + O_2 \rightarrow 2H_2O$

$2 \times 2.016 = 4.032$ g; 32 g; $2 \times (2.106 + 16) = 36.032$

3 g of $H_2$ require $O_2 = 32/4.032 \times 3$ g $= 23.8$ g

Thus, $O_2$ (29 g) is present in excess. Hence, $H_2$ is limiting reactant.

$H_2O$ formed = $36.032/4.032 \times 3$ g $= 26.8$ g

$O_2$ left unreacted = $29 - 23.8 = 5.2$ g.

$= 3.7$ kg

By mole concept , Mass of 22.4 litres of $CO_2$ at STP = Molecular mass in grams

$= 12 + 2 \times 16 = 44$ g

To calculate the volume and mass of $CO_2$ formed from 1 litre of $O_2$ at STP.

Applying Gay Lussac's Law of Gaseous volumes, 1vol. $O_2$ produces $CO_2 = 2$ vol.

1 litre of $O_2$ will produce $CO_2 = 2$ litres at STP

**Que 19 – One litre of oxygen at STP is made to react with three litres of carbon monoxide at STP. Calculate the mass of each substance found after the reaction. Which one is the limiting reactant?**

Ans – The chemical equation representing the reaction is :

$2CO + O_2 \rightarrow 2CO_2$

2 vol.1 vol.2 vol.

Applying Gay Lussac's Law of Gaseous Volumes, 1 vol. of $O_2$ reacts with CO = 2 vol.

1 Litre of $O_2$ will react with CO = 2 litres at STP

But the volume of CO taken = 3 litre at STP

Volume of CO (at STP) left unused = 3 – 2 = 1 litre.

Now by mole concept , mass of 22.4 litres of CO at STP= Molecular mass in grams

= 12 + 16 = 28 g

Mass of 1 litre of CO at STP = 28/22.4 x 1 = 1.25 g

Thus, the mass of CO left unused = 1.25 g

**Que 20 – Ammonia contains 82.35% of nitrogen and 17.65% of hydrogen. Water contains 88.90% of oxygen and 11.10% of hydrogen. Nitrogen trioxide contains 63.15% of oxygen and 36.65% of nitrogen. Show that these data illustrate the law of reciprocal proportions.**

Ans- In $NH_3$, 17.65 g H combine with N = 82.35 g

Therefore 1 g of H combine with N = 82.35/17.65 = 4.67 g

In $H_2O$, 11.10 g of H combine with O = 88.90 g

Therefore 1 g of H combine with O = 88.90/ 11.10 = 8.01 g

Therefore ratio of the masses of N and O which combine with fixed mass(=1g) of H

= 4.67 : 8.01 = 1 : 1.72

In $N_2O_3$, ratio of masses of N and O which combine with each other

= 36.85 : 63.15 = 1 : 1.71

Thus, the two ratios are same. Hence, it illustrate the law of reciprocal proportions.

Mass of 2 litres of $CO_2$ at STP = 44/22.4 x 2 = 3.928 g

Thus , the mass of $CO_2$ produced = 3.928 g

As oxygen has been completely used up , hence oxygen is limiting reactant

**Que 21 – Calculate the mass of an atom of silver.**

Ans- 1 mole of Ag atoms = 108 g

$= 6.022 \times 10^{23}$ atoms

$6.022 \times 10^{23}$ of silver have mass = 108 g

∴ Mass of one atom of silver = $108/ 6.022 \times 10^{23}$

$= 1.793 \times 10^{-22}$ g.

**Que 22 – How many atoms and molecules of sulphur are present in 64.0 g of sulphur ($S_8$)?**

Ans – Molecular formula of sulphur = $S_8$

Molecular mass of sulphur = 32 x 8 = 256.0 u

1 mole of sulphur molecules = 256 g of sulphur contain $6.022 \times 10^{23}$ molecules of sulphur

Now, 256 g of sulphur contain $6.022 \times 10^{23}$ molecules

∴ 64 g of sulphur will contain = $6.022 \times 10^{23}$ x 64/256

$= 1.506 \times 10^{23}$ molecules

1 molecule of sulphur ($S_8$) contains atoms of sulphur

∴ $1.506 \times 10^{23}$ molecules of sulphur will contain sulphur atoms = $8 \times 1.506 \times 10^{23}$

$= 1.2048 \times 10^{24}$ atoms

**Que 23 – Calculate the number of molecules present in 350 $cm^3$ of $NH_3$ gas at 273 K and 2 atmosphere pressure.**

Ans– Given conditions At STP

$V_1 = 350\ cm^3$ $V_2 = ?$

$T_1 = 273$ K $T_2 = 273$ K

$P_1 = 2$ atmospheres $P_2 = 1$ atm

Applying gas equation : $P_1V_1/T_1 = P_2V_2/T_2$,

We get 350 x 2/ 273 = 1 x V2/273 or V2 = 350 x 2 / 273 x 273/1 = 700$cm^3$

By mole concept , 1 mole of NH3 = 6.022 x $10^{23}$ molecules

= 22400 $cm^3$ at STP

Thus, 22400 $cm^3$ of NH3 at STP contain 6.022 x $10^{23}$ molecules

∴ 700 $cm^3$ of NH3 at STP will contain = 6.022 x $10^{23}$ / 22400 x 700 = 1.882 x $10^{22}$ molecules

Alternatively, applying gas equation = PV x nRT

n = PV/ RT

= 2 atm x 0.350 L/ 0.0821 L atm $K^{-1}$ $mol^{-1}$ x 273 K

= 0.0312 mole

∴ No. of molecules = 0.0312 x 6.022 x $10^{23}$ = 1.88 x $10^{22}$

**Que 24 – Two bulbs B1 and B2 of equal capacity contain 10g oxygen (O2) and ozone (O3) respectively. Which bulb will have greater number of O- atoms and which will have greater number of molecules?**

Ans – 10 g O2 = 10/32 mole = 10/32 x 6.02 x $10^{23}$ molecules

= 1.88 x $10^{23}$ molecules

= 2 x 1.88 x $10^{23}$

= 3.76 x $10^{23}$ atoms

10 g O3 = 10/48 x 6.02 x $10^{23}$ molecules

= 1.254 x $10^{23}$ molecules

= 3 x 1.254 x $10^{23}$ atoms

= 3.76 x $10^{23}$ atoms

Thus, both contain the same number of atoms but bulb B1 contains more number of molecules

**Que 25 – Calculate the volume of hydrogen liberated at STP when 500 $cm^3$ of 0.5 N sulphuric acid reacts with excess of zinc ( H = 1, O = 16, S=32).**

Ans – The chemical equation representing the reaction is :

Zn + H2SO4 → Zn SO4 + H2

98.0 g → 22400 $cm^3$ at STP

To calculate the amount of H2SO4 in 500 $cm^3$ of 0.5 N H2SO4 solution.

Strength/ litre of H2SO4 solution = Normality x Eq. mass = 0.5 x 49 = 24.5 g/litre

Now, 1000 $cm^3$ of the acid solution contain 24.5 g of pure H2SO4

∴ 500 $cm^3$ of the acid solution will contain pure H2SO4 = 24.5 x 500/1000 = 12.25 g

To calculate the volume of H2 liberated at STP

98 g of H2SO4 react to liberate 22400 $cm^3$ of H2 at STP

∴ 12.25 g of H2SO4would liberate H2 at STP = 22400/98 x 12.25 $cm^3$ = 2800 $cm^3$

Thus, the volume of hydrogen liberated at STP = 2800 $cm^3$

**Que 26 – What volume of oxygen at STP is required to affect complete combustion of 200 $cm^3$ of acetylene and what would be the volume of carbon dioxide formed?**

Ans – The chemical equation representing the combustion of acetylene is :

2 C2H2 + 5 O2 → 4 CO2 + 2 H2O

2 vol.5 vol.4 vol.

To calculate the volume of O2 at STP required to affect complete combustion of 200 $cm^3$ of acetylene. Applying Gay Lussac's Law of gaseous volumes.

2vol. Of C2H2 require O2 for complete combustion = 5 vol.

∴ 200 $cm^3$ of C2H2 will require O2 for complete combustion = 5/2 x 200

= 500 $cm^3$ at STP

Thus, the volume of O2 required = 500 $cm^3$ at STP

To calculate the volume of CO2 produced at STP

Applying Gay Lussac's Law of gaseous volumes, 2 vol. of C2H2 produce CO2 = 4 vol.

∴ 200 $cm^3$ of C2H2 at STP will produce CO2 = 4/2 x 200 = 400 $cm^3$ at STP

Thus, the volume of CO2 produced = 400 $cm^3$ at STP.

**Que 27 – Calculate the mass of a molecule of carbon dioxide.**

Ans – 1 mole of CO2 = 44 g

= 6.022 x $10^{23}$ molecules

Thus, 6.022 x $10^{23}$ molecules of CO2 have mass = 44 g

∴ 1 molecule of CO2 has mass = 44/ 6.022 x $10^{-23}$ = 44 x $10^{-23}$ / 6.022 g

= 7.307 x $10^{-23}$ g

**Que 28 – Calculate the number of molecules present (i) in 34.20 grams of cane sugar (ii) in one drop of water having mass 0.05g.**

Ans – (i) – 1 mole of C12H22O11 = 342 g

= 6.022 x $10^{23}$

Now, 342 g of cane sugar contain 6.022 x $10^{23}$ molecules

∴ 34.2 g of cane sugar will contain = 6.022 x $10^{23}$ / 342 x 34.2 = 6.022 x $10^{22}$ molecules

(ii) 1 mole of H2O = 18 g = 6.022 x $10^{23}$ molecules

Mass of one drop of water = 0.05 g

Now, 18 g of H2O contain = 6.022 x $10^{23}$ molecules

∴ 0.05 g of H2O will contain = 6.022 x $10^{23}$ / 18 x 0.05 = 1.673 x $10^{21}$ molecules.

**Que 29 – (i) Assuming the density if water to be 1g/$cm^3$, calculate the volume occupied by one molecule of water.**

**(ii) Assuming the water molecule to be spherical, calculate the diameter of the water molecule.**

Ans – (i) 1mole of H2O = 18 g = 18 $cm^{23}$

= 6.022 x $10^{23}$ molecules of H2O

Thus, 6.022 x $10^{23}$ molecules of H2O have volume = 18 $cm^3$

∴ 1 molecule of H2O will have volume = 18/ 6.022 x $10^{23}$ $cm^3$

= 2.989 x $10^{-23}$ $cm^3$

(ii) As water molecule is assumed to be spherical , if R is its radius, then its volume will be

4/3 $\pi R^3$ = 2.989 x $10^{-23}$ $cm^3$ or $R^3$ = 7.133 x $10^{-24}$

Take n = ( 7.133)$^{1/3}$ ∴ log n = 1/3 log 7.133 = 1/3 x 0.8533 = 0.2844

n = Antilog 0.2844 = 1.925

∴diameter of water molecule = 2 x 1.925 x $10^{-8}$ cm = 3.85 x $10^{-8}$ cm

**Que 30 – The density of 3 M solution of NaCl is 1.25 g $mL^{-1}$. Calculate the molality of the solution.**
Ans – Molality = Moles of the solute/ Mass of the solvent in kg

3 M NaCl solution means that 3 moles of NaCl are present in 1 L of the solution.

Mass of 1 L , i.e., 1000 mL of the solution = volume x density = 1000 mL x 1.25 g $mL^{-1}$ = 1250 g

Mass of 3 moles of solute (NaCl) = 3 mol x 58.5 g $mol^{-1}$ = 175.5 g

∴ Mass of solvent = 1250 – 175.5 g = 1074.5 g = 1.0745 kg

∴ Molality of solution = 3 mol/ 1.0745 = 2.79 mol $kg^{-1}$ = 2.79 m

**Que 31 – Calculate the mass of (i) 0.1 mole of KNO3 (ii) 1 x $10^{23}$ molecules of methane**
Ans – (i) 1 mole of KNO3 = 101 g

∴ 0.1 mole of KNO3 = 101 x 0.1 = 10.1 g of KNO3

(ii) 1 mole of CH4 = 16 g = 6.022 x $10^{23}$ molecules

i.e., 6.022 x $10^{23}$ molecules of methane have mass = 16 g

∴ 1 x $10^{23}$ molecules of methane will have mass = 16/ 6.022 x $10^{23}$ = 2.657 g

**Que 32 – How many grams of NaOH should be dissolved to make 100 $cm^3$ of 0.15 M NaOH solution?**
Ans – 1000 $cm^3$ of 0.15 M NaOH = 0.15 mole

∴ 100 $cm^3$ of 0.15 M NaOH contain NaOH = 0.15/ 1000 x 100

= 0.015 mole = 0.015 x 40 g

= 0.6 g

**Que 32 – Calculate the molarity of water if its density is 1000 kg/$m^3$.**
Ans – Molarity of water means number of moles of water in 1 litre of water

1 L of water = 1000 $cm^3$ = 1000 g

1000 g H2O = 1000/18 moles = 55.56 moles

∴ Molarity of water = 55.56 mol $L^{-1}$